GGG-S-58E
May 25, 1983
SUPERSEDING
GGG-S-58D
June 20, 1971

FEDERAL SPECIFICATION

SAW, RADIAL-OVERARM, WOODWORKING

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers radial-overarm, electric-motor-driven circular, woodworking saws.

1.1.1 Federal Specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification. The saws shall be of the following styles, and sizes, as specified (see 6.2):

Style A - With legs.
Style B - Without legs (for portable use).
Size 10 - Standard duty (5/8-inch arbor), 2-3/4-inch depth of cut, 10 inch max. blade diameter.
Size 12 - Standard duty (5/8-inch arbor), 3-3/4-inch depth of cut, 12 inch max. blade diameter.
Size 14 - Heavy duty (1-inch arbor), 3-3/4-inch depth of cut, 14 inch max. blade diameter.
Size 16 - Heavy duty (1-inch arbor), 4-1/2-inch depth of cut, 16 inch max. blade diameter.
Size 18 - Heavy duty (1-inch arbor), 5-3/4 inch depth of cut, 18 inch max. blade diameter.
Size 18L - Heavy duty (1-inch arbor), 5-3/4 inch depth of cut (extra long over-arm), 18 inch max. blade diameter.

FSC 3220

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification

GGG-B-440 - Blade, Circular Saw, Woodcutting.

Federal Standard

FED-STD-123 - Marking for Shipment (Civil Agencies).
FED-STD-H28 - Screw Thread Standards for Federal Services.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Description from established distribution points in their agencies.)

Military Specifications

MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-M-18058 - Machinery, Metal and Woodworking, Preparation for Delivery of.

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MS35792 - Handle, Bail-Chest, Recessed.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.1.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

Occupational Safety and Health Act of 1970 (OSHA):

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and Amendments.
Occupational Safety and Health Administration Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

National Electrical Manufacturers Association (NEMA):

MG 1 - Motors and Generators.
ICS 1 - Industrial Controls and Systems, General Standards For.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

National Fire Protection Association (NFPA):

No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

3. REQUIREMENTS

3.1 Description. The saws shall consist of the following: (1) A non tilting, horizontal frame with worktable; (2) an upright column supporting a horizontal nonretractable overarm attached to or supported by the frame and extending above the table top, with the arm having integral machine tracks for traverse of the cutting head; and (3) a yoke supporting a cutting head consisting of a revolving nonportable electrical motor with saw blade.

3.1.1 Performance. The saws shall be capable of performing straight line cuts of the sizes specified in Table I. The cuts performed shall be accurate within the following limits:

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Straightness - 0.003 inch per foot.
Squareness - 0.005 inch per foot.
45 mitering - 0.010 inch per foot.
45 beveling - 0.010 inch per foot.

3.1.2 Size. Each saw size shall be capable of using the maximum diameter of saw blade specified in Table I.

3.2 First article (preproduction model). Unless otherwise specified (see 6.2), the contractor shall furnish a saw for examination and test within the time frame specified (see 6.2) to prove, prior to starting production, that his production methods and choice of design detail will produce saws that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Material not specified shall be new and of a quality necessary to produce saws to meet the requirements specified herein. Materials used shall be corrosion resistant or provided with a coating that provides protection against the various forms of material deterioration in the applicable operating and storage environments to which they may be exposed.

3.3.1 Galvanic Corrosion. Dissimilar metals as defined in MIL-STD-889 shall not be in intimate contact without suitable protection in order to minimize or circumvent galvanic corrosion.

3.4 Safety. All exposed parts which are energized electrically shall be insulated, fully enclosed or guarded. All moving parts which are hazardous to the operator shall be guarded in accordance with OSHA requirements, Subpart O, Paragraph 1910.212 and 1910.213. Protective devices shall not impair the operating functions. All hand-operated devices such as adjusting levers, latches, locks, handwheels, and cranks shall be located away from the cutting tool.

3.5 Ease of maintenance. All major assemblies, installed attachments, and parts subject to wear shall be accessible for adjustment, repair, and replacement.

3.5.1 Common parts. Common hardware and fastening devices such as bolts, nuts, screws, and similar items shall be held to a minimum number of types and sizes and shall be installed with means to prevent loss of tightness without swaging, peening, or staking. All new screw threads shall be threaded to the standards of FED-STD-H28.

3.6 Frame. The frame shall support the saw and have the radial-overarm assembly attached. The frame shall be a metal structure, braced so that the saw may be operated at maximum output without visible vibration or deflection of the overarm.

3.7 Table work-top assembly. The table work-top assembly shall consist of the table work-top and front guide fence and may be equipped with a rear guide fence.

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3.7.1 Table work-top. The table work-top shall be a nontilting surface. The table work-top for Size 10 and 12 saws shall consist of 3/4 inch minimum thickness resin bonded plywood with hardboard surfaces, or particle board; and Size 14, 16, 18, and 18L saws shall consist of 1 inch minimum thickness resin bonded plywood with hardboard surfaces, particle board, edge grained fir, or vertical grained fir. The table work-top shall be of sufficient depth to give support to lumber of sizes as specified in Table I. When specified (see 6.2), the table work-top, with permanent or removable extensions, shall be of a size that when the saw is at the limit of saw travel toward the operator, when crosscutting or when positioned for full capacity normal ripping, no part of the saw teeth or cutting edge will project beyond the front edge or ends of the table work-top.

3.7.2 Front guide fence. The front guide fence shall be dressed white pine or other soft-short grain wood or particle board of not less than 3/4-inch thickness. The fence shall extend not less than 3/4 inch above and to both ends of the tabletop. The fence shall be removable. A single guide fence may be furnished to serve as both front and rear guide fence (see 3.7.3).

3.7.3 Rear guide fence. A rear guide fence, if required for maximum ripping capacity, may be fastened to the rear of the table work-top or may be removable. The fence shall be of dressed white pine or other soft short-grain wood or particle board of not less than 3/4-inch thickness. The fence shall extend not less than 3/4 inch above and to both ends of the tabletop.

3.8 Radial-overarm assembly. The radial-overarm assembly shall consist of an upright column supporting a horizontal arm. The radial-overarm assembly shall be adjustable for mitering at least 45 degree on one side of the normal crosscutting position and at least 60 degree on the opposite side of the normal crosscutting position as shown in Figure 1. The cutting head, with mountings and motor, shall be assembled to the radial-overarm assembly. The cutting head shall move freely on precision ball or roller bearings, rolling on fully machined tracks. The radial-overarm assembly shall be equipped with bearings, stops, miter locator, miter lock, miter scale, rip lock, and rip scale in accordance with the following requirements. All scales and pointers shall be adjustable to provide calibration.

3.8.1 Bearings. Bearings for the radial-overarm assembly shall be of the precision ball or roller radial type, prelubricated, and sealed or shielded for life. Bearings shall withstand without damage all radial loads and thrust loads produced when the saws are operating as specified herein.

3.8.2 Stops. The radial-overarm assembly shall be equipped with stops at the travel limits of the cutting head. The limiting stop shall be cushioned to prevent damage to the saw. The rear stop shall prevent the blade or accessories from contacting the base or column during any operation shown in Figure 2. The radial-overarm assembly shall be prevented from leaving the arm in the event of failure of a limiting stop or arm end cap, when the push button switch is an integral part of the end cap.

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3.8.3 Miter locator. The radial-overarm assembly locator shall locate and maintain the accuracy of the saw when performing mitering operations at the 45 degree positions.

3.8.4 Miter lock. A positive holding, quick-acting, miter lock shall lock the radial-overarm assembly anywhere within the mitering range of the saw.

3.8.5 Miter scale. The radial-overarm assembly shall be equipped with a miter scale or calibrated dial. The scale shall be calibrated in increments of not less than 1 degree for the complete mitering range of the saw. Starting from zero at least every 15th degree shall be numbered and every 5th degree shall be indicated by a longer division line.

3.8.6 Rip lock. A positive-holding, quick-operating rip lock shall lock the radial-overarm assembly anywhere within the ripping range of the saw. The rip lock shall not distort any part of the saw it contacts.

3.8.7 Rip scale. The radial-overarm assembly shall be equipped with one or more rip scales to indicate the width of rip. The scales shall be lithographed, etched, or engraved and shall be graduated at 1/16 inch intervals. The 1-inch calibrations shall be numbered.

3.8.8 Rotation direction. The guard or motor assembly shall be permanently marked to show the correct arbor rotation direction. All arbor-mounted accessories shall be permanently marked with the correct rotation direction on the side installed away from the motor.

3.9 Cutting head with yoke and motor. The cutting head shall consist of the saw guard and antikickback device, arbor mountings, and the motor. The cutting head shall be mounted in a yoke for traveling on or with the radial-overarm assembly. The cutting head, in combination with the radial-overarm assembly, shall permit the positioning of the cutting tools to perform any of the operations specified herein. When tested as specified in 4.5.2.7, the total cutting head deflection shall not exceed that specified in Table II and the dial indicator shall return to within 0.001 inch of zero in each case.

TABLE II. Cutting Head Deflection.

Saw size	Load, pounds	Total deflection, inch maximum
10	2	0.010
12	3	.020
14	7-1/2	.012
16	10	.010
18	15	.010
18L	15	.015

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The cutting head shall have a horizontal radial range of travel of not less than 180 degree when measured from a normal left-hand crosscutting position (see Figure 1) to either inside or outside the ripping position.

3.9.1 Bevel locator. The saw shall be equipped with a bevel locator consisting of a positive-locating, quick-acting device or devices with means of adjustment for locating the cutting head in not less than three positions. The normal horizontal crosscutting position, the 45 degree beveling position, and the normal vertical position.

3.9.2 Bevel lock. The saw shall be equipped with a positive-holding, quick-operating bevel lock for locking the cutting head anywhere within the beveling range of the saw.

3.9.3 Bevel scale. The saw shall be equipped with a bevel scale or calibrated dial. The scale shall be calibrated in increments not greater than 1 degree for the complete beveling range of the machine. Starting from zero, every 5th-degree calibration shall be longer than the intermediate calibrations.

3.9.4 Cutting-head locator. The cutting head shall be equipped with a locator consisting of a positive-locating, quick-acting device or devices with means of adjustment for locating the cutting head in at least three positions: The normal crosscutting position, the inside rip position, and outside rip position. A clamp shall be provided to insure tightness under vibration.

3.9.5 Adjustments. Adjusting mechanisms shall be provided at each point where it is necessary to change the position of the saw to perform any operation shown in Figure 2. The adjustments shall be made without the use of shims. The adjusting mechanisms shall maintain their setting during full load operation and shall remain accurate until the adjustment is reset. All frequently used adjusting mechanisms shall be provided with a means to facilitate quick and accurate settings. Positive-adjusting mechanisms shall be provided at all necessary points to permit the saw to be adjusted for accurate cuts initially and to compensate for wear which might later affect the accuracy of the saw. Saw-adjustment mechanisms shall include the following means for adjustment:

a. Either tabletop to be parallel to line of saw travel or line of saw travel to be parallel to tabletop for all types of cutting operations.
b. Sawing head to make the saw blade perpendicular to tabletop.
c. Line of saw travel to be perpendicular to guide fence.
d. Extreme rear position of the cutting head by means of a stop to prevent the saw blade, cutting head, or guard from striking the upright column when making any cut shown in Figure 2.
e. Bevel, miter, and rip scale so that their readings will be coincident with stops or saw kerf.
f. Ways or rollers to eliminate cutting-head play.
g. Motor orientation so that the trailing edge of the blade follows the leading edge through the cut kerf (heel adjustment).

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The saw shall have a vertical adjustment to provide an unobstructed clear distance between the bottom of the motor and the worktable in accordance with Table III.

TABLE III. Vertical Adjustment.

Saw Size	Vertical clear distance (motor bottom to table), inches minimum
10	6
12	9-1/2
14	13
16	14-1/2
18	14-1/2
18L	14-1/2

The vertical adjustment shall be accomplished by a protected screw mechanism. One revolution of the crank or handwheel shall raise or lower the cutting head 1/16 or 1/8 inch. The crank, handwheel, or adjacent area shall be marked to show elevation rotation direction. The screw shall turn without undue force but the construction shall be such that the accuracy of setting will be maintained during at least 1/2 hour of actual continuous crosscut operations using the maximum depth and width of cut specified in Table I. Mechanical means shall be provided for adjusting and maintaining the accuracy of alinement of each saw for every usable position.

3.9.6 Saw guard, antikickback device, and splitter. Each saw shall be equipped with an adjustable saw guard which shall protect the operator under all operating conditions specified herein. The upper part of the saw guard shall cover not less than one-half of the sawblade periphery down to a point that will include the end <#f the saw arbor. The saw guard shall be equipped with a reversible dust spout to throw the dust away from the operator during all sawing operations. The lower part of the saw guard shall be either: (1) A type that completely covers the lower periphery of the saw blade on both sides and raises as the blade enters the cut and drops back into position upon completion of the operation; or (2) a clamptype guard that shall clamp to the upper guard and shall be fully adjustable to cover all portions of the cutting tool above the material being cut. The lower saw guard shall operate freely and smoothly while performing any of the cuts shown in Figure 2. An antikickback device shall be furnished to protect the operator during bevel and straight ripping operations and shall be adjustable to any thickness of stock within the capacity of the saw with which it is used. The saw shall be provided with a roll-or knife-type adjustable splitter to prevent material from pinching the blade.

3.9.7 Arbor. The motor shall have a round, solid arbor which is integral with the rotor. The arbor shall be of sufficient length to mount the maximum width dado cutters as shown in Table I. The diameter of the arbor at the saw blade

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mounting position shall be in accordance with Table I. Eccentricity of the arbor and flange at the saw blade mounting position shall not exceed 0.002 inch (total indicator reading) and 0.0025 inch (total indicator reading), respectively.

3.9.8 Mountings. Each cutting head shall be equipped with a motor-mounting yoke. The yoke shall be constructed and balanced so as to maintain its setting accurately in any position while the saw is operating.

3.9.9 Automatic return device. The saw shall be equipped with an automatic return device that will return the cutting head to the retracted position when released by the operator. The device shall return and hold the cutting head in full retracted position when released from any position of its travel on the overarm.

3.10 Gears, worms, and worm wheels. Gears, worms, and worm wheels, if used, shall provide smooth transmission of power and speed within standard gear design practices.

3.11 Electrical equipment.

3.11.1 Motor types. The motor size and type shall be selected from Tables I and IV, as specified (see 6.2).

3.11.2 Motors. Motors shall be of the continuous duty induction type, shall be as specified in Table IV, and shall conform to NEMA Standard MG1 except for frame size, mounting dimensions, and shaft and starting torque. Motors shall be the direct-drive type equipped with precision ball or roller bearings of the radial-thrust type sealed, shielded, and lubricated for life. The motor armature or roter and shaft shall be balanced dynamically.

3.11.3 Motor protection and wiring. Motor protection and wiring shall be in accordance with NFPA Standard No. 70.

3.11.4 Starter. The saws shall be equipped with a pushbutton starter of the type specified in Table IV. Unless otherwise specified herein, the starter shall be mounted in a NEMA Standard ICS 1, Type I enclosure. When weatherproof construction is specified (see 6.2), (the starters shall be installed in a location to provide weatherproof protection; or shall be mounted in a NEMA Standard ICS 1 Type IV enclosure, at the option of the contractor). The stop-start pushbutton shall be located within sight and easy reach of the operator in any operating position of the saw. A wiring diagram shall be secured to the inside of the removable starter cover.

3.11.5 Cord. Leads between the motor and switch shall conform to Type SJ, SO, SJT, S, or ST "Hard Service'' Cord of NFPA Standard No. 70.

3.11.6 Power brake. A power brake as specified in Table IV shall be incorporated in the motor control circuit. With the saw operating at full no-load-speeds, the power brake shall automatically stop the saw blade motor

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within the applicable braking time range specified in Table V. The brake shall decelerate the saw blade smoothly to a stop and stop the saw blade motor within the time range specified in Table V without overheating or without loosening of the saw blade nut. After the power brake is operated, the motor shall restart without resetting or adjusting the brake.

TABLE V. Braking Time Range for Power Brakes.

Saw size	Braking time, seconds maximum
12	25
14	18
16	20
18	45
18L	45

3.12 Additional equipment. All additional equipment shall be as specified herein and shall operate on the size of saw specified. Saw blades and cutters shall fit the arbor without the use of bushings. Saw blades except those used for dadoing shall be of a diameter capable of cutting the depths specified.

3.12.1 Equipment furnished with the saw. The following equipment shall be furnished with each saw:

1 each - combination saw blade.
1 each - set of tools normally furnished by the contractor.

3.12.2 Optional equipment. When specified (see 6.2), any or all of the following equipment shall be furnished with each saw:

Set of dadoing equipment.
Set of dado spacing collars.
Set of shaping cutters (size 10 and 12 only).
Rafter notching head (size 10 and 12 only).
Power ripfeed (sizes 16, 18, and 18L only).
Power crossfeed (sizes 16, 18, and 18L only).
Manual chain crossfeed (sizes 16, 18, and 18L only).
Blades, circular saw, woodcutting of sizes and types specified. Set of roller conveyor tables with legs (size 14, 16, 18, and 18L only).

3.12.2.1 Dadoing equipment. Dadoing equipment shall consist of a dado head and a guard when the guard specified in 3.9.6 is not of sufficient width to cover the dado head. The dado head shall consist of two (outside) hollow-ground saw blades and a sufficient number of (inside) dadoing cutters to perform any dadoing cut from a minimum width of 1/8 inch and each width in the 1/16-inch increments up to and including the dado head width specified in Table I. The minimum

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diameter of the dado head shall be in accordance with Table VI. For widths in excess of Table I requirements, a rafter notching head equipped with dadoing knives as specified in 3.12.2.3 shall be used.

Table VI. Dado Heads.

Saw size	Dado head	
	Minimum diameter (inches)	Minimum depth of cut (inches)
10	6	1
12	6	1
14	8	1-1/2
16	10	2
18	12	3
18L	12	3

3.12.2.1.1 Spacing collars. Spacing collars for dadoing equipment shall consist of one each of 1/16, 1/8, and 1/4 inch in thickness. The spacing collar hole shall correctly fit the arbors specified in Table I and shall be of sufficient outside diameter to support the cutter heads.

3.12.2.2 Shaping cutters. The set of shaping cutters shall be of the head and inserted-blade type. Cutter shapes shall be as specified (see 6.2).

3.12.2.3 Rafter notching head. The rafter notching head shall be of the size as specified (see 6.2).

3.12.2.4 Power ripfeed. The power ripfeed shall provide for adjustable rates of feed from 30 to 75 feet per minute (fpm). The power ripfeed shall be provided with a motor (and switch) having sufficient horsepower to feed lumber (both hardwood and softwood) through the machine when set up and in operation for ripping. The feed wheels and mechanisms shall not mar the finished work when lumber being ripped is held against the guide fence and the specified rates of feed are maintained during ripping operations. The material f#ed wheel or sprocket shall be guarded. The power ripfeed shall be adjustable so that it may be positioned to prevent interference for crosscutting and other variety cutting.

3.12.2.5 Power crossfeed. The power crossfeed shall provide for adjustable rates of forward feed from 0 to not less than 40 fpm. The crossfeed shall be provided with controls for forward and backward movement of the cutting head, and a master stop control which on actuation shall stop the cutting head feed and rotation at any position of its travel. The crosscut capacity of the saw with power crossfeed installed shall not be reduced by more than 1 inch. If the crossfeed requires a power source other than the saw motor the power source shall be either standard shop air line furnishing 50 pounds per square inch (psi) of air pressure or electrical service 110 to 120, AC 60 Hertz.

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3.12.2.6 Manual chain crossfeed. The chain crossfeed shall be manually operated by a removable crank or handwheel located near the front of the saw. An adjustment shall remove all backlash between the cutting head and the crank or handwheel. The manually-operated chain crossfeed shall move the cutting head through its normal traverse length while the saw is cutting at its normal cutting rate.

3.12.2.7 Blades. Blades shall conform to GGG-B-440.

3.12.2.8 Roller conveyor tables. The set of roller conveyor tables shall consist of two tables 8 feet long plus or minus 1 inch, each having not less than five metal rollers not less than 12 inches long for size 14 saws and not less than 18 inches long on sizes 16, 18, and 18L and not less than 1.875 inches in diameter (see 3.18 for trailer-mounted saws). The metal rollers shall be supported by and shall rotate on ball or roller bearings. The saw table and roller table shall be provided with means for securely fastening their inboard ends to the ends of the saw table at an adjustable height which will permit lumber placed on the conveyor table to just clear the saw table. The outboard ends of each conveyor table shall be supported by two legs braced and having individual, positive-locking adjustments for height variations up to 6 inches. When attached to the saw table, the conveyor table with its legs shall support a 20-foot hardwood timber, 8 inches square, without perceptible vibration or sidesway.

3.13 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.14 Fungus control. When specified (see 6.2), fungusproofing shall be in accordance with commercial practices for equipment going to the tropics.

3.15 Identification marking. Each saw shall be legibly and permanently marked with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined. The marking shall include the type or model designation and any other descriptive data necessary to identify spare or replacement parts.

3.16 Identification plate. When specified (see 6.2), each saw shall be provided with a separate identification plate conforming to MIL-P-514, Type I, Style 2, Composition C of Type I, Grade A, Class 1 material. The plate shall contain the following information marked with 3/16-inch high characters:

Production equipment code number (to be supplied in contract). Activity identification (to be inscribed by the receiving activity).

3.17 Treatment and painting. Unless otherwise specified (see 6.2), the saws shall be treated and Painted in accordance with the suppliers standard practice.

3.18 Military requirements. When trailer-mounted, size 16 saws are specified (see 6.2 and 6.6), the requirements specified in 3.18.1 through 3.18.4 shall apply.

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3.18.1 Dimensions and weight. The overall dimensions and weight of the size 16 saw, shall not exceed the following:

width	50 inches
depth	35 inches
height	45 inches
width of roller conveyor table	12 inches
length of roller conveyor table	5 feet
weight	750 pounds

The width shall be measured from side-to-side. The depth shall be measured from the extreme rear of the table frame to the front of the table frame.

3.18.2 Saw modifications. The saw shall be modified to provide it with rollers to permit the saw to be rolled from the traveling position to the operating position and back without binding. The general configuration of the rollers shall be as shown in Figures 3 and 4. Holddowns shall be provided to prevent shifting of the saw while in transit on the trailer and while the saw is being operated on the tailgate. The general configuration of the holddowns shall be as shown in Figures 5 and 6. Two handles conforming to MS35792-1 shall be mounted in the front of the saw base 6 inches = 1 inch from each corner for pulling the saw from the trailer onto the tailgate.

3.18.3 Power ripfeeds. Power ripfeeds will not be provided.

3.18.4 Elevating handle. The elevating handle shall be located on the overarm at or near the front at the overarm; or the handle must extend outward from the base at least 8 inches when being used. When attached to the base and not in use, the handle shall not interfere with operator's movement around saw nor interfere with the trailer. The handle shall be permanently attached to the saw.

3.19 Style A, with legs or cabinet. The Style A saws shall be equipped with legs of steel or cast iron. The legs shall be constructed and braced to provide a rigid support for the saw. When specified (see 6.2), a metal cabinet with door and latch shall be provided for Style A, size 10 saws only.

3.20 Style B, without legs (for portable use). The Style B saws shall be furnished without legs for bench and portable use.

3.21 Workmanship. The saws shall be free from defects such as incomplete welds, rust, cracks, or other defects that could impair the operation or serviceability of the saws.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in

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the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. Preproduction inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The saw shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The saw shall be tested as specified in 4.5.2.1 through 4.5.2.11, as applicable. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, Inspection Level II.

4.4.2 Examination. Each saw shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Test. Each saw selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.4 and 4.5.2.8 and the AQL shall be 2.5.

4.5 Inspection procedure.

4.5.1 Examination. The saw shall be examined for the following defects:

101. Dimensions not as specified.
102. Assembly incomplete or incorrect.
103. Parts, components, tools, or equipment missing or not functioning as required.
104. Safety devices and guards missing or not as specified; sharp corners and sharp edges not rounded off.
105. Material not as specified.
106. Workmanship not as specified.
107. Treatment or painting not as specified.
108. Accessories, identification or other marking missing.

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109. Parts subject to wear not accessible.
110. Adjustments not as specified.
111. Saw blade diameter not as specified.
112. Return device not as specified.

4.5.2 Tests. The cutting head and horizontal arm shall be adjusted prior to conducting the tests specified herein.

4.5.2.1 Arbor and flange. Check the eccentricity of the saw arbor at the point where the cutting tools will be applied. Check the eccentricity of the flange against which the saw blade is clamped. Runout in excess of 0.002 inch (total indicator reading) for the arbor and 0.0025 inch (total indicator reading) for the flange shall constitute failure of this test.

4.5.2.2 Cutting-head travel. Check the accuracy of the cutting-head travel by (1) crosscutting a board equal to the crosscut capacity specified in table I, (2) holding a steel straight-edge along the edge cut, and (3) checking with a feeler gage vertically between the straight-edge and cut board. Inaccuracy in straightness in excess of 0.003 inch per foot shall constitute failure of this test.

4.5.2.3 Crosscut test. Check the accuracy of saw travel and guide fence by (1) crosscutting a board with the end to be checked to the left of the saw; (2) crosscutting with the other end of the board to the right of the saw; (3) checking each cut by placing it in a steel square with a known squareness accuracy of not less than 0.001 in 10 inches, with the fence edge of the board held to the inside of the square; and (4) checking with a feeler gage vertically between the cut edge and the steel square. Width of board to be cut shall be as specified in Table I under miter capacity. Inaccuracy in squareness in excess of 0.005 inch per foot shall constitute failure of this test.

4.5.2.4 Mitering. Check the accuracy of the 45 degrees miter by (1) mitering two boards not less than 20 inches long by 6 inches nominal width; (2) placing the miters together and checking with a carpenter's steel square along the inside edges; and (3) checking with a feeler gage vertically at the mating surfaces. Inaccuracy in excess of 0.010 inch per foot per board (0.020 inch per foot total) shall constitute failure of this test.

4.5.2.5 Beveling. Perform the beveling test as specified in 4.5.2.4 except set the saw for 45 degrees beveling instead of 45 degrees mitering.

4.5.2.6 Shaping. Make a 5/16-inch bend and a 5/16-inch cove cut 11/16-inch wide and 11/16-inch deep around a rectangular test piece. The test piece shall be maple, birch, beech, walnut, or hardwood of similar cutting qualities. Chatter marks, gouges, roughness, or other markings shall constitute failure of this test.

4.5.2.7 Cutting-head deflection. Upon completion of the tests specified in 4.5.2.1 through 4.5.2.6 and without any adjustment, test the saw for deflection as follows. Position the cutting-head at its maximum distance from the upright column and clamp securely. Lock the cutting head in the crosscut position. Use

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a spring scale to apply the load specified in Table II to the saw arbor axially in one direction. Measure the deflection of the arbor. Repeat the procedure in the other direction. Determine the total deflection of the saw arbor. Lock the cutting head in the rip position at its maximum distance from the upright column. Apply the load specified in Table II vertically to the arbor both up and down. Determine the total deflection of the saw arbor with respect to the tabletop. When the load is removed, the dial indicator should return to within 0.001 inch of zero in each case. Deflection greater than that specified in Table II or failure to return to within 0.001 inch of zero shall constitute failure of this test.

4.5.2.8 Power brake. Test the braking time of the saw by operating the cutting-head motor at normal operating speed. Shut off the motor and check the time required for the motor to come to a complete stop. Braking time not within the range specified in Table V shall constitute failure of this test.

4.5.2.9 Power ripfeed. When a power ripfeed is specified, test the saw as follows. Install roller conveyor tables on the saw. Move the saw back out of the cutting position and test the rate of feed of the power ripfeed at 30 and 75 fpm, feeding lumber selected by the contractor. Then rip, oak or maple, at least one-half the maximum depth of cut specified in Table I. Inability of the ripfeed to feed lumber at 30 and 75 fpm or failure to rip the above specified lumber one-half the depth specified in Table I shall constitute failure of this test.

4.5.2.10 Power crossfeed. When a power crossfeed is specified, test the saw as follows: Operate the power crossfeed (not sawing lumber) at 40 fpm and at three different slower speeds. During this operation set the saw for its full crosscut travel and not less than three intermediate settings. Stop the cutting head by actuating the master stop. Restart the saw and make 10 cuts across a 2 by 6-inch oak or maple plank. During this operation, the saw shall be set to feed at approximately 10 fpm and at a travel sufficient to clear the plank by approximately 1 inch. Inability of the saw to (1) operate at 40 fpm or intermediate speed, or (2) stop the cutting head travel when the master stop is actuated shall constitute failure of this test.

4.5.2.11 Chain crossfeed. When a chain crossfeed is specified, test the saw as follows: Traverse the cutting head (not sawing) with the chain crossfeed through its normal traverse length. Retract the cutting head and cut off a 2- by 6-inch oak or maple plank. Inability of the cutting head to travel its normal traverse length or backlash between the cutting head and crank or handwheel shall constitute failure of this test.

4.6 Inspection of preparation for delivery. The preservation, packaging, packing, and marking for Level A or B shall be examined and tested in accordance with the Quality Assurance Provisions of MIL-M-18058. Commercial preservation, packaging, packing and marking shall be examined to determine compliance with MIL-STD-1188.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or commercial as specified (see 6.2).

5.1.1 Level A. Each saw with accessories, attachments, repair parts, tools, and publications shall be preserved and packaged in accordance with the Level A requirements of MIL-M-18058.

5.1.2 Commercial. Each saw with accessories, attachments, repair parts, tools, and publications shall be preserved and packaged in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Saws, preserved as specified in 5.1, shall be packed in accordance with the Level A requirements of MIL-M-18058.

5.2.2 Level B. Saws, preserved as specified in 5.1, shall be packed in accordance with the Level B requirements of MIL-M-18058.

5.2.3 Commercial. Saws, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Civil agencies. Packages and shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military activities. Marking for Level A or B shall be in accordance with MIL-STD-129. Marking for commercial shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The saws are intended for use in woodworking and cabinet shops, as well as in mobile shops, and are particularly adaptable for use in boxing and crating work in shipping departments.

6.1.1 Standard lumber size coverage. Table VII is provided to show the classes of saws required for specific thicknesses of lumber.

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TABLE VII. Saw Capacities for Standard Lumber.

Saw size	Rough size of standard lumber (inches)	Finished size of standard lumber (inches)
10	2	1-1/2
12	2 and 4	1-1/2 and 3-1/2
14	2 and 4	1-1/2 and 3-1/2
16	2 and 4	1-1/2 and 3-1/2
18	2, 4 and 6	1-1/2, 3-1/2, and 5-1/2
18L	2, 4 and 6	1-1/2, 3-1/2, and 5-1/2

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Style, and size required (see 1.2).
c. When preproduction model is not required (see 3.2).
d. Time frame required for submission of preproduction model (see 3.2).
e. When tabletop depth is to be greater than the forward position of the saw (3.7.1).
f. Size and type of motor required (see 3.11.1).
g. When weatherproof construction is required (see 3.11.4).
h. Optional equipment required (see 3.12.2).
i. Cutter shapes required (see 3.12.2.2).
j. Size of rafter notching head required (see 3.12.2.3).
k. When fungusproofing is required and type of fungusproofing (see 3.14).
l. When a separate identification plate is required (see 3.16).
m. Treatment and painting required if other than as specified (see 3.17).
n. When trailer-mounted, size 16 saws are required (see 3.18).
o. When a metal cabinet with door and latch is required (see 3.19).
p. Levels of preservation, packaging, and level of packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production saws from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish saws conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, foundation and anchor bolt plans, wiring diagrams, and contractor's maintenance and operation manual to be furnished with each saw.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of saws.

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6.6 When the saw is to be used for trailer mounting in SC-3220-97-CL-E01, the requirements in 3.18 through 3.18.4 should be specified.

6.7 Classification changes. The relationship between the saw sizes in GGG-S-58D and the size designations in this issue is as follows:

GGG-S-58D	GGG-S-58E
1	10
2	12
Deleted	Deleted
3	14
Deleted	Deleted
4	16
Deleted	Deleted
Deleted	Deleted
5A	18
5B	18L

6.8 Information figures. Figures 1 and 2 are descriptive and not restrictive and are not intended to preclude the furnishing of saws which conform to this specification.

MILITARY INTERESTS:

Custodians:
Army - ME
Navy - SH
Air Force - 99

Review activities:
Navy - YD
Air Force - 84
DLA - GS

User activity:
Navy - CG

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

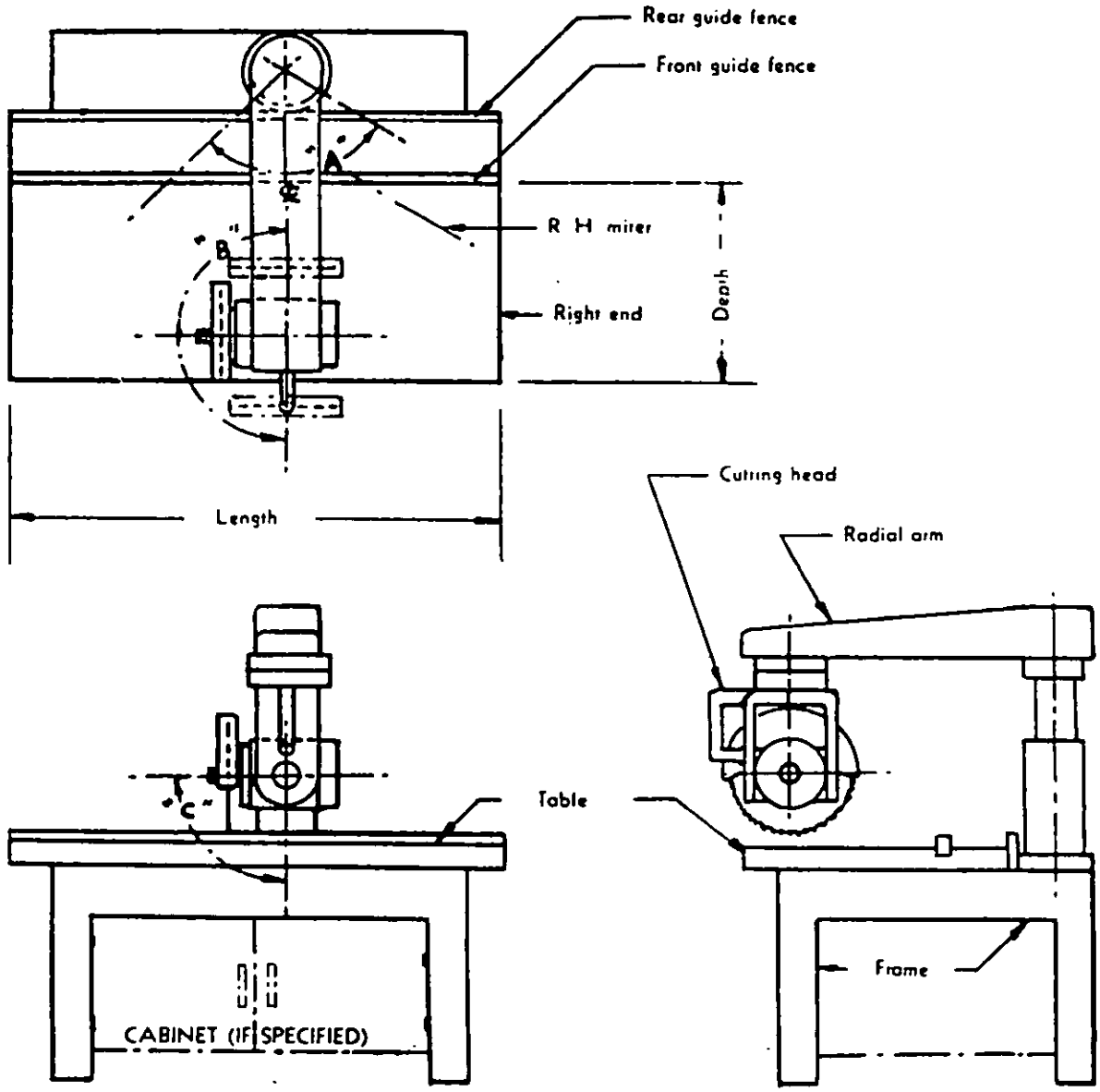
PREPARING ACTIVITY:
ARMY - ME

Project 3220-0134

U.S. GOVERNMENT PRINTING OFFICE : 1983 - 381-595/5399

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"A" = 105° Mitering { 45° One side of centerline. 60° Opposite side of centerline.
"B" = 180° Cross-cutting to right- and left-hand ripping.
"C" = 90° Beveling.

Figure 1. Saw adjustments (minimum).

X848

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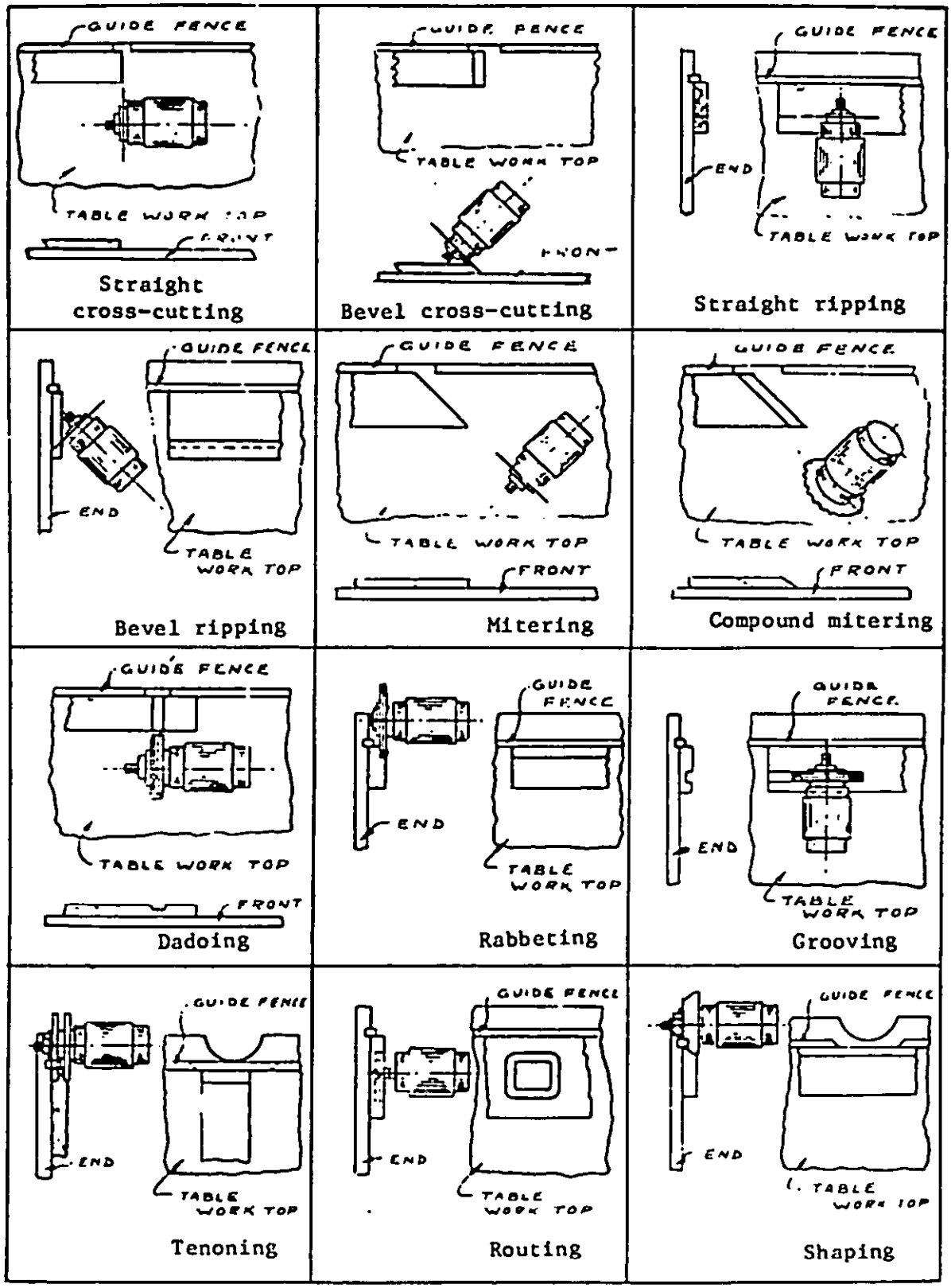


Figure 2. Typical saw operating shown without safety devices.

X849

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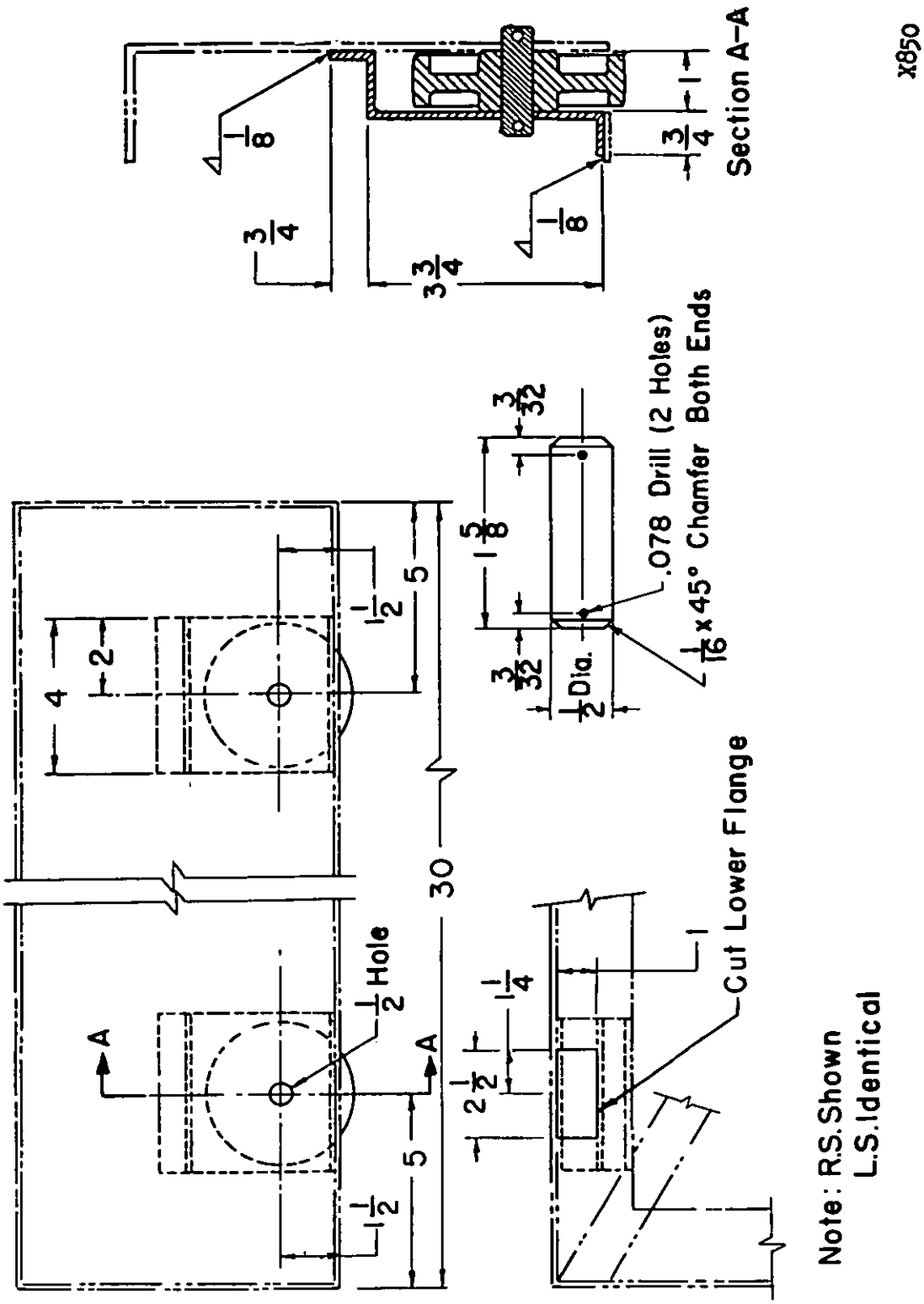


Figure 3. - Roller Bracket

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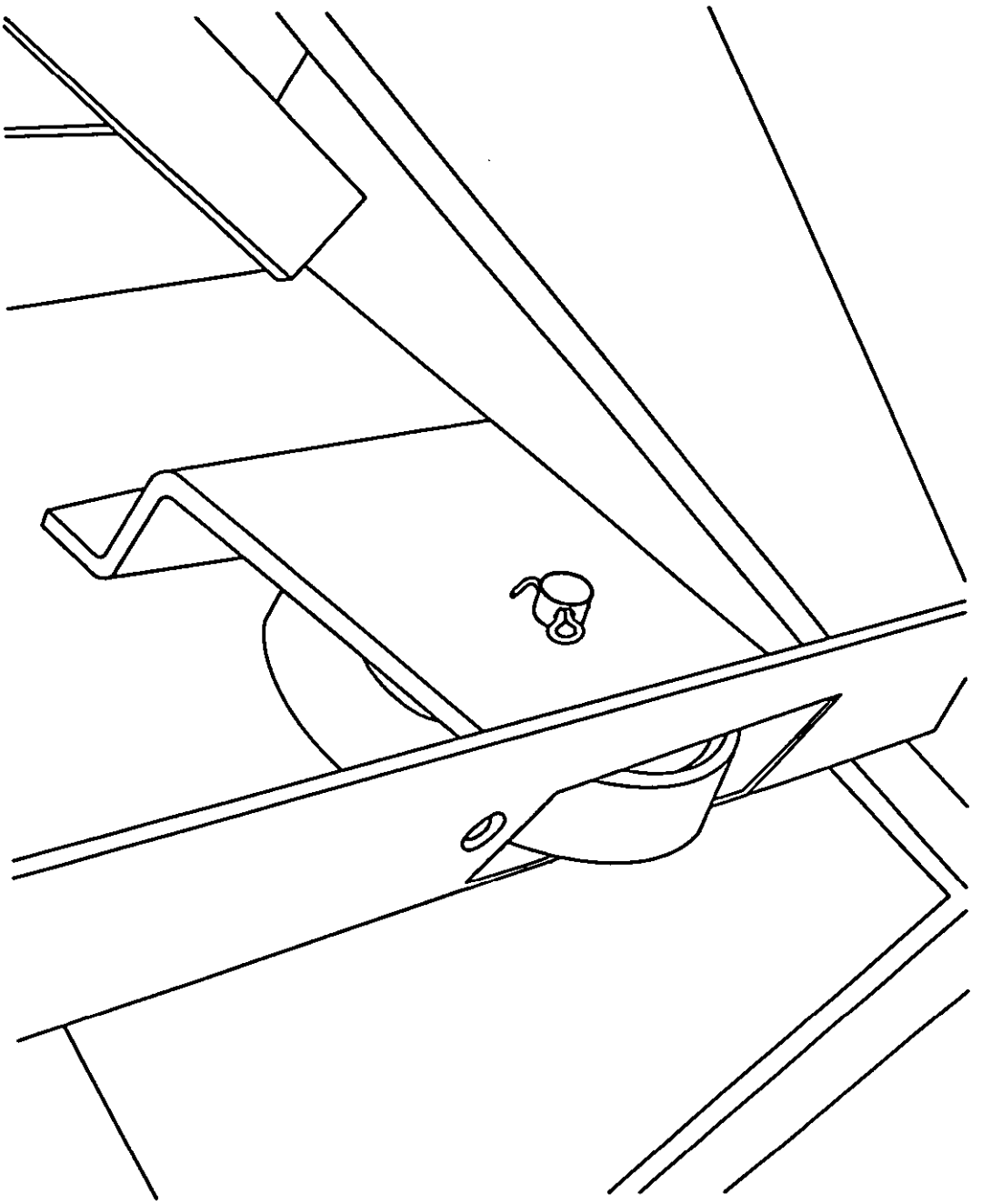


FIGURE 4. ROLLER,SAW, TABLE FRAME X851

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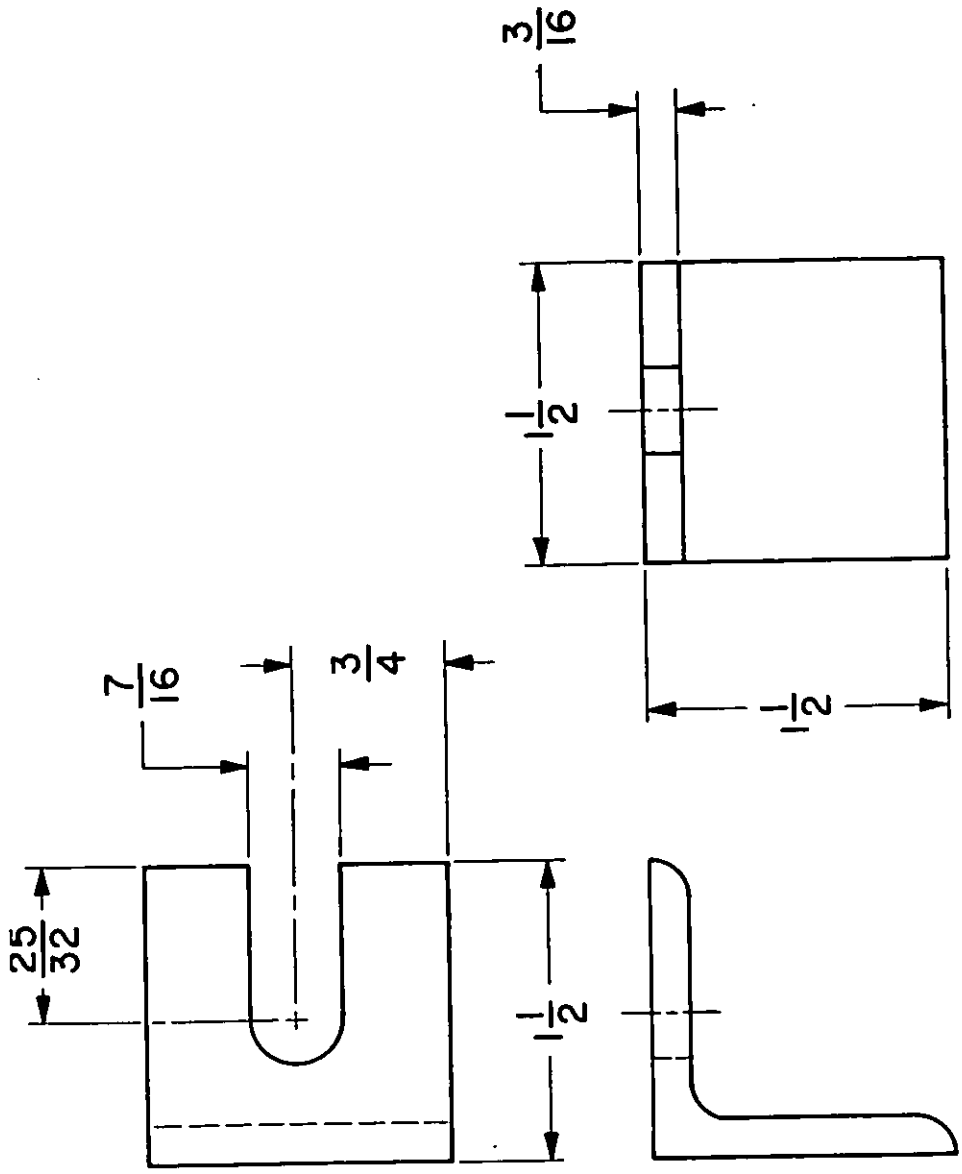


FIGURE 5. HOLD DOWN BRACKET CX-1001

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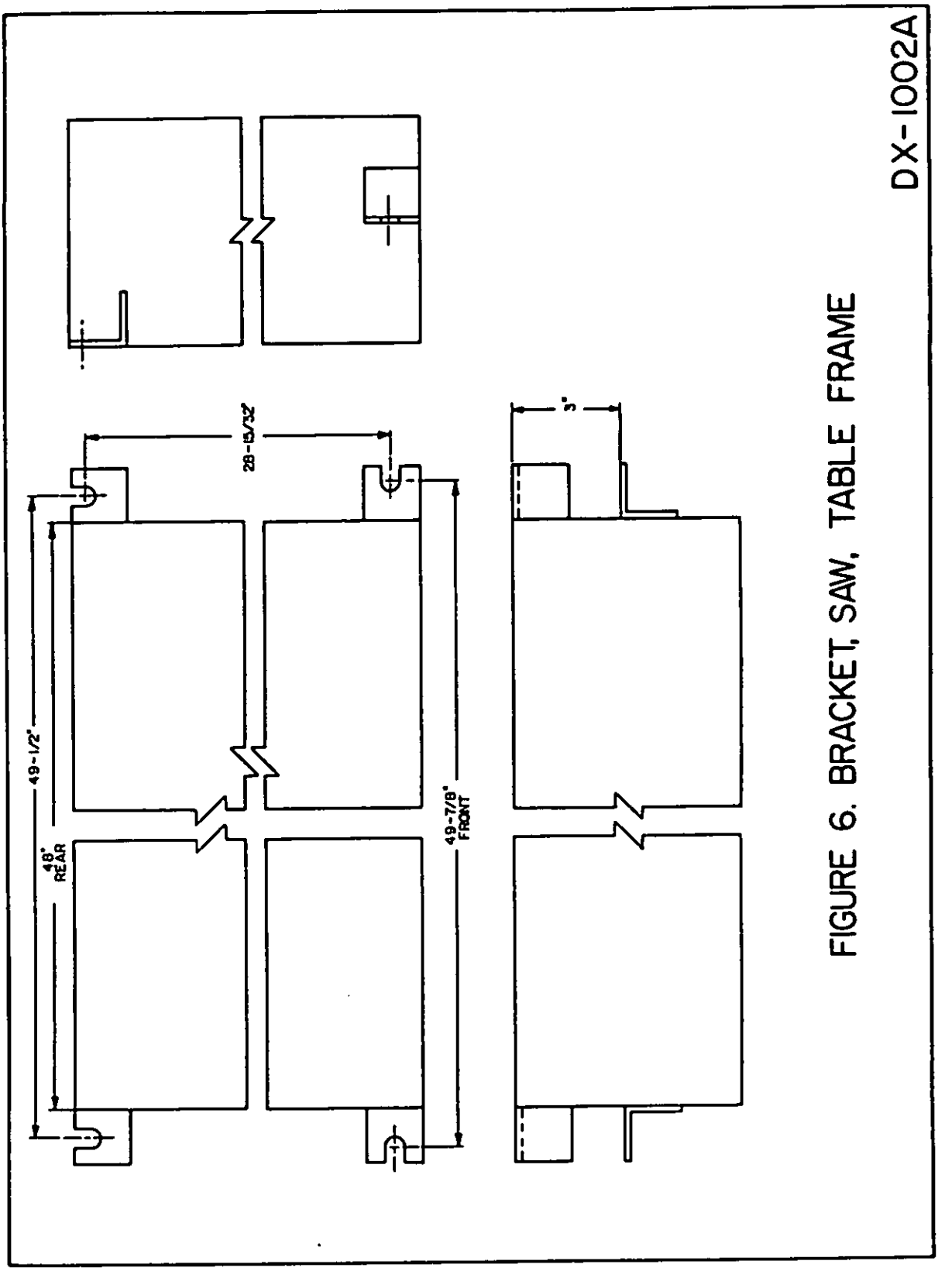


FIGURE 6. BRACKET, SAW, TABLE FRAME

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3. DOCUMENT TITLE
Saw, Radial-Overarm, Woodworking

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (If applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Geraldine F. Harlow	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3469 (2) AUTOVON 654-3468
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: SATBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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